Kaggle/petfinder

Group: Team A

1. Task Definition and Dataset Overview

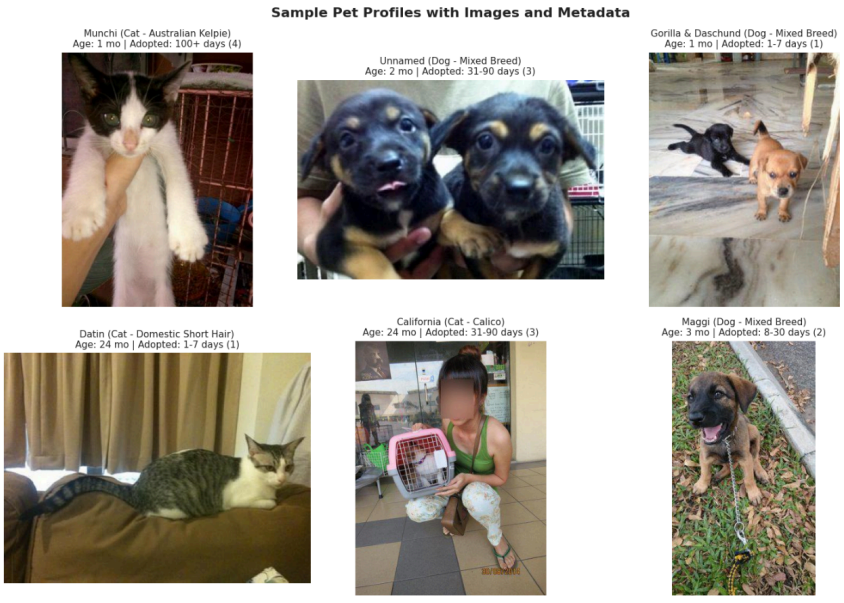
Task Type Definition:

- This is a **Multiclass Classification** task.
- **Why:** The objective is to predict the `AdoptionSpeed` of listed pets. The target variable consists of five discrete category labels (0: adopted on the same day, 1: 1-7 days, 2: 8-30 days, 3: 31-90 days, 4: not adopted after 100 days). Even though these labels represent ordered timeframes, the model must classify each pet into one of these specific discrete categories rather than predicting a continuous numeric value (which would be the case for regression).

Dataset Presentation:

The primary tabular dataset contains 14,993 rows and 24 columns (including the target) for training, and 3,972 rows and 23 columns for testing. Beyond the tabular data, the full dataset also includes optional images, Google Vision image metadata, and Google Natural Language sentiment/entity outputs.

What & How: This data represents the real-world challenge of placing shelter animals into homes. Each row represents a PetFinder listing/profile (identified by a `PetID`), which can represent a single pet or a group of pets. The dataset combines numerical, categorical, and text fields (such as age, breed, health, fee, and profile descriptions) to evaluate which characteristics most strongly influence how quickly a pet is adopted. The primary source is the official Kaggle competition page: [PetFinder Adoption Prediction]. The original data was provided by **PetFinder.my**, Malaysia's leading pet adoption platform.



2. Feature Descriptions

Feature	Type	Meaning	Relevance
Type	Categorical (binary)	Animal type: 1 = Dog, 2 = Cat.	Adoption patterns may differ between dogs and cats.
Name	Text	Name of the pet; may be missing.	Presence or content of a name may relate to listing quality.
Age	Numerical	Age of the pet in months.	Younger and older pets may have different adoption rates.
Breed1	Categorical (nominal)	Primary breed, encoded by BreedID; names in breed_labels.csv.	Different breeds may have different adoption patterns.
Breed2	Categorical (nominal)	Secondary breed, if applicable; 0 = none.	Provides information about mixed breeds.
Gender	Categorical (nominal)	1 = Male; 2 = Female; 3 = Mixed (group listing).	Adoption preferences may differ by gender.
Color1	Categorical (nominal)	Primary coat color, encoded by ColorID; names in color_labels.csv.	Coat appearance may affect adopter preferences.
Color2	Categorical (nominal)	Secondary coat color; 0 = none.	Describes additional coat coloration.
Color3	Categorical (nominal)	Third coat color; 0 = none.	Describes additional coat coloration.
MaturitySize	Categorical (ordinal)	Expected adult size: 1 = Small; 2 = Medium; 3 = Large; 4 = Extra Large; 0 = Not specified.	Pet size may influence adoption preferences.
FurLength	Categorical (ordinal)	Fur length: 1 = Short; 2 = Medium; 3 = Long; 0 = Not specified.	Fur length may affect care requirements.
Vaccinated	Categorical (nominal)	Vaccination status: 1 = Yes; 2 = No; 3 = Not sure.	Vaccination may influence adoption decisions.
Dewormed	Categorical (nominal)	Deworming status: 1 = Yes; 2 = No; 3 = Not sure.	Indicates preventive healthcare.
Sterilized	Categorical (nominal)	Sterilization status: 1 = Yes; 2 = No; 3 = Not sure.	May influence adopter preferences.
Health	Categorical (ordinal)	Health: 1 = Healthy; 2 = Minor injury; 3 = Serious injury; 0 = Not specified.	Health condition may affect adoption speed.
Quantity	Numerical (discrete)	Number of pets represented in the listing.	Group listings may differ from individual listings.
Fee	Numerical	Adoption fee in Malaysian Ringgit (MYR); 0 = Free.	Higher fees may influence adoption decisions.
State	Categorical (nominal)	Malaysian state, encoded by StateID; names in state_labels.csv.	Adoption activity may vary by geographic region.
RescuerID	Identifier	Unique identifier of the rescuer or organization.	May capture differences between rescuers.
VideoAmt	Numerical (discrete)	Number of uploaded videos.	Videos may provide useful information to adopters.
Description	Text	Written profile description of the pet.	Text may describe personality, health and adoption details.
PetID	Identifier	Unique identifier of the pet adoption listing.	Used to identify listings; not a direct pet characteristic.
PhotoAmt	Numerical (discrete)	Number of uploaded photos.	Photos may improve listing visibility and attractiveness.
AdoptionSpeed	Target (ordinal)	Adoption speed: 0 = same day; 1 = 1-7 days; 2 = 8-30 days; 3 = 31-90 days; 4 = not adopted after 100 days.	Target variable to predict.

Note: Breed, coat color, and state are represented by numerical codes, with their meanings provided in the corresponding label files. PetID and RescuerID are identifiers, while AdoptionSpeed is the target variable.

3. Data Quality Assessment

We assessed data quality by examining missing values, categorical code validity, cross-column consistency, duplicates, identifiers, and numerical outliers. Issues were identified using Pandas and the dataset's official label files. The `clean_petfinder()` function was used to correct selected inconsistencies without removing any records.

Missing Values and Data Consistency

Finding	Affected records	Decision / Reason
Missing Name (NaN)	1,265 (8.4%)	Keep; valid unnamed pets
Placeholder names	151 (1.0%)	Convert to NaN
Missing Description	13 (0.1%)	Replace with empty string
Unknown health-related status	1,868 / 1,781 / 1,815	Keep as categorical values
Invalid Breed1 = 0	5 (0.03%)	Use valid Breed2 where applicable
Breed2 = Breed1	1,510 (10.1%)	Set Breed2 to 0
Cat assigned a dog breed	12 (0.1%)	Keep; uncertain correction
Group listings (Quantity > 1)	3,428 (22.9%)	Keep; valid groups
Identical except PetID	16 (0.1%)	Keep; potentially distinct pets
PhotoAmt stored as float	All records	Convert to integer

Only `Name` and `Description` contain actual NaN values. Unknown health-related statuses are encoded as separate categories and are not detected by `isna()`. Considering both types of unknown information, 24.5% of listings contain at least one unknown value.

Duplicates and Identifiers

No duplicate `PetID` values were found. The dataset contains 5,595 unique rescuers, and no `RescuerID` appears in both the training and test sets. This suggests a rescuer-disjoint split, making group-based validation using `RescuerID` worth considering to reduce potential data leakage.

Outliers and Numerical Distributions

Numerical features were examined using descriptive statistics and the interquartile range (IQR) method.

	`Age` (months)	`Quantity`	`Fee`	`VideoAmt`	`PhotoAmt`
Median / Q3 / max	3 / 12 / 255	1 / 1 / 20	0 / 0 / 3000	0 / 0 / 8	3 / 5 / 30
Above IQR fence	10.0%	22.9%	15.5%	3.8%	6.2%

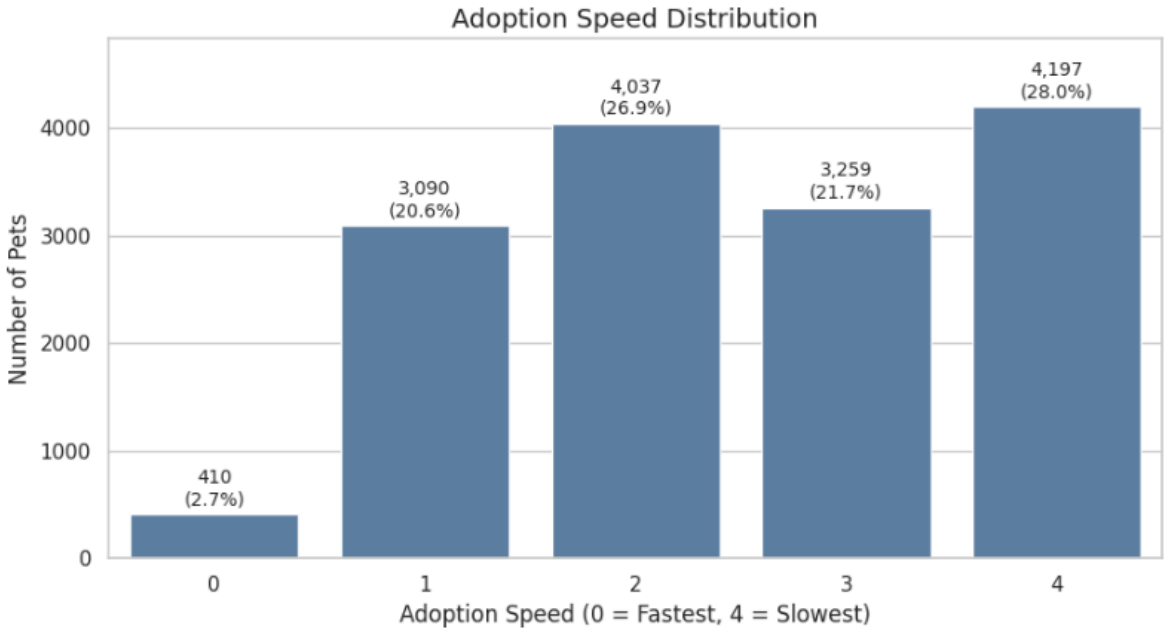
The IQR method can overestimate outliers for discrete and zero-heavy features such as `Quantity`, `Fee`, and `VideoAmt`. Age is strongly right-skewed, with several unusually old animals. However, extreme observations were retained because they may represent valid records rather than errors.

Overall Data Quality

The dataset is generally suitable for classification, although it contains missing textual information, unknown categorical values, and some inconsistent breed codes. Selected issues were corrected consistently in training and test data, without removing rows or modifying the target variable. Further preprocessing may include categorical encoding and transformations of skewed numerical features before modeling.

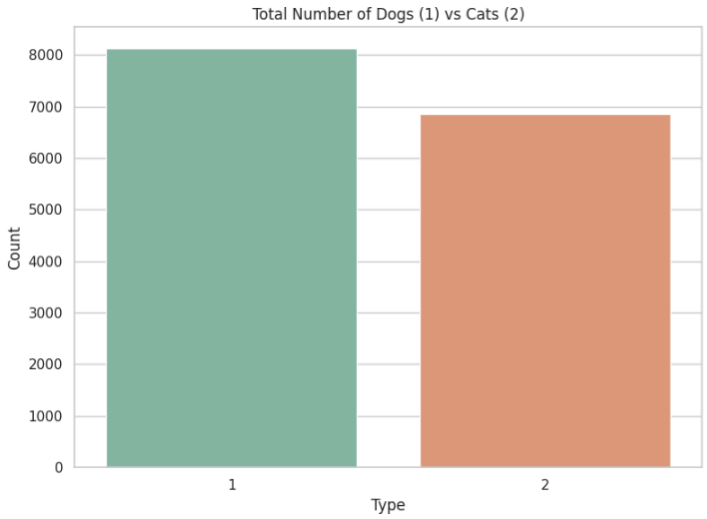
4. Exploratory Data Analysis & Visualizations

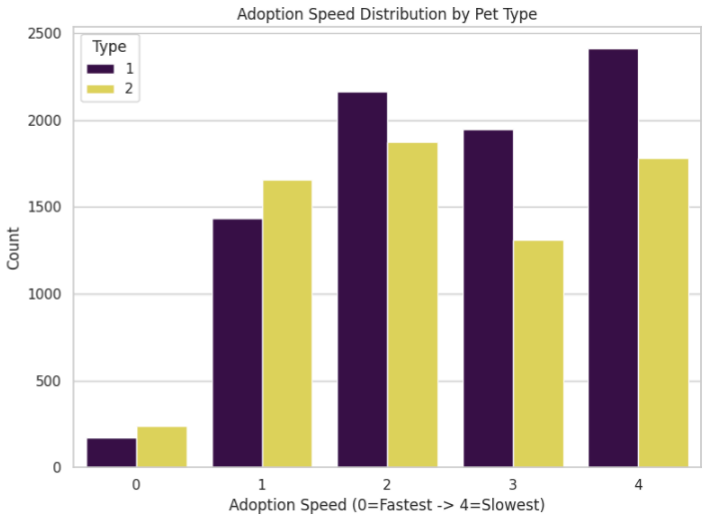
Exploratory data analysis was performed using Pandas, Matplotlib, and Seaborn to examine feature distributions and their relationships with AdoptionSpeed.



Adoption Speed Distribution

Category 4 is the most frequent (4,197 listings; 28.0%), while category 0 is the least frequent (410; 2.7%). This class imbalance should be considered when training and evaluating classification models.



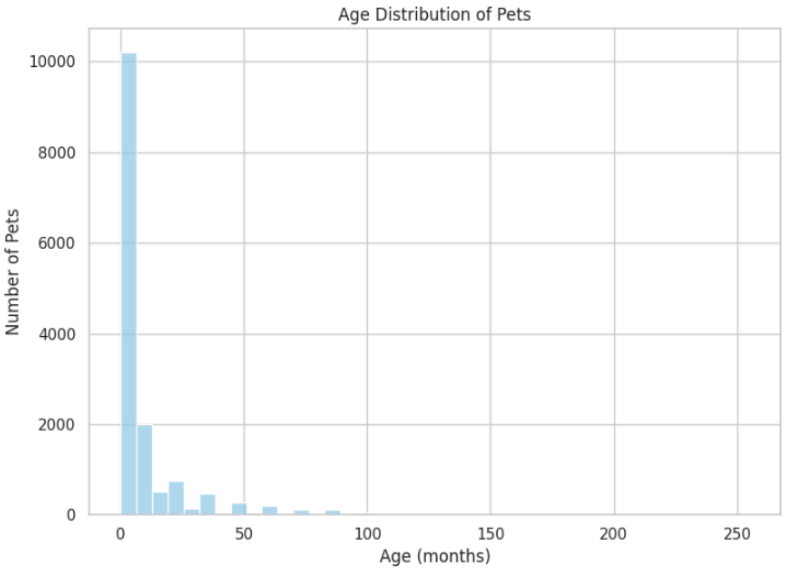


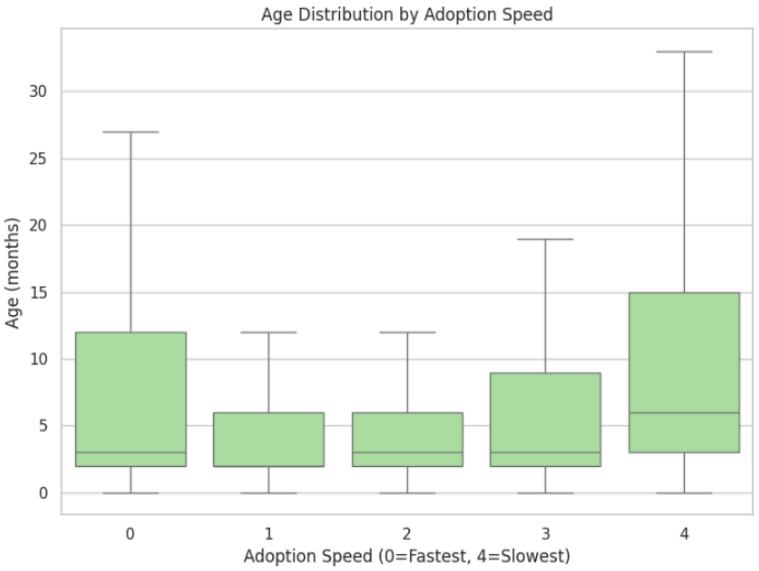
Distribution of Pet Types

Dogs represent approximately 54% of listings, while cats account for 46%. The dataset is therefore relatively balanced by animal type, with slightly more dogs than cats.

Adoption Speed by Pet Type

Cats have more listings in the faster adoption categories (0–1), while dogs have more in categories 2–4. However, these are absolute counts, and the dataset contains more dogs overall. A percentage-based comparison would be needed to draw reliable conclusions about differences in adoption speed.





Age Distribution of Pets

The histogram shows a strongly right-skewed age distribution, with most listings concentrated at younger ages. A small number of unusually old animals are also present, indicating potential outliers.

Age Distribution by Adoption Speed

Category 4 has a higher median age (approximately 6 months) than categories 1–3 (approximately 3 months). This suggests that age may be associated with adoption speed, although the distributions overlap considerably.

Key Takeaways

The analysis identified a class imbalance in AdoptionSpeed, a relatively balanced distribution of dogs and cats, and a strongly right-skewed age distribution. The higher median age in slower adoption categories suggests that age may be a useful predictor. Further analysis is needed to evaluate the relationship between animal type and adoption speed.